## Welcome to Mastering STRATEGY

### Part 4 of 12

**This is the fourth part of Mastering Strategy, a weekly series which will be published in all worldwide editions of the newspaper every Monday up to and including December 13th.**
**The aim of the series is to give readers an overview of this important business topic, to describe popular techniques for implementing business strategy, and to discuss the latest ideas from some of the world's leading strategic thinkers. In the process we hope it makes you a better and more informed manager.**

● Part Four today continues our examination of price competition which began with a discussion of game theory in Part Three (see FT Oct 11). Price wars may be great for the consumer but as Luis Garicano and Rob Gertner explain on this and the next two pages they can seriously damage a company's profits. Mindful of the practical difficulties and regulatory sensitivities of co-ordination the authors set out a number of the strategies which can result in what they call "implicitly co-operative outcomes". The prisoner's dilemma case below the picture – based on behaviour in the First World War trenches – adds richness to their insights.

● On pp 6-7 the focus switches to alliance strategy – a topic introduced in Part Two (FT Oct 4) with a piece urging managers carefully to compare the virtues of acquisitions and joint ventures before proceeding down a particular path. Today Will Mitchell emphasises not only the importance of pre-alliance planning but of the need to learn from the alliance experience and to develop relevant corporate capabilities. He recommends companies to set up a special unit dedicated to alliance management.

● Noel Tichy in Part Three (FT Oct 11) has already underlined the imperative of growth strategies in today's markets – a topic tackled today by his University of Michigan colleague Aneel Karnani on pp 8 and 10. Karnani sets out five possible growth "directions" and analyses the advantages and disadvantages of each. He comes to the conclusion that market penetration, globalisation and forward integration are currently the most promising.

● On the whole there is no shortage of new ideas - the challenge for everyone is how to emulate the late Akio Morita of Sony who was so good at turning them into commercial winners. On pp 12 and 13 Ron Adner throws light on this dilemma arguing that a key aspect of the innovation selection process is the setting of management expectations.

● Finally in this Part John Kay (author of the stimulating introduction to our series in Part One, FT Sep 27) considers the special challenge of running a regulated business. Contrary to the view of many outsiders this is no easy task: most of the conventional problems are there but in addition regulated companies have to become expert in government and regulatory relationships as well as understand a particular set of competitive dynamics.

**Mastering Strategy, regular readers already know, brings together four powerful and distinguished writing teams in the area. The collective wisdom and advice of top flight professors and other faculty from the University of Michigan Business School, the University of Chicago Graduate School of Business, Insead (based near Paris, France) and Oxford University's Saïd School of Business in the UK represents a rich mix of traditions and outlooks and a rare synthesis of North American and European views.**

**The full contents of the series – published thematically in Part Two – are shown chronologically on page 15. Please also note the signposts to related articles and ideas for additional reading in the margins.**
**As with previous FT Masterings we welcome feedback and letters which, space permitting, we will endeavour to publish.**

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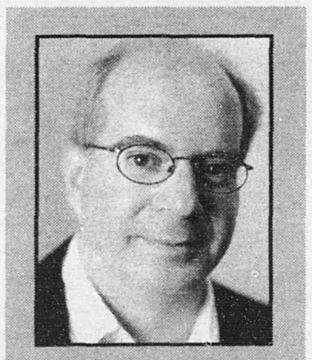
# The dynamics of price competition

## Summary

Pricing has an important strategic role in economics but the dynamics of price competition is frequently misunderstood. **Luis Garicano** and **Robert Gertner** focus here on the incentives companies have to use price as a competitive weapon, the situations in which price cutting can pay off, and the conditions which lead to higher prices being charged by all companies in an industry. Changing the rules of the game so as to avoid price wars can raise antitrust issues and needs to be approached with caution, the authors warn. They explain why price setting in future is likely to be buffeted by the ease with which customers can indulge in comparative shopping on the internet.



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Nothing wipes out profits like price competition. Companies in industries like personal computer retailing and book selling earn little money, largely because of intense price competition. Despite its strategic importance and central place in economics, it remains difficult to predict differences in price competition across industries and over time within industries. There are no simple, robust rules. Yes, price competition is on average less intense in more concentrated industries, but the relationship is surprisingly weak. There are many concentrated industries – like commercial aircraft – where price competition is intense just as there are un-concentrated industries that have limited price competition, credit cards for example.

In most industries, companies choose the prices they sell at. Of course, this is constrained by the actions of competitors and consumers, but nonetheless it is a choice. Yet, the press often refers to the "outbreak of a price war" as though it was a contagious disease rather than the result of conscious decisions by companies believing they are acting in their own interests. The interesting reality, as we shall explain, is that price wars often lower or destroy industry profits.

## Price wars

The logic of price competition was first captured in a simple model by nineteenth century French economist Joseph Bertrand. The model is as follows. Suppose that your company and another company are the only producers of a certain good. There is only one potential client, who would be willing to pay up to $50 per unit for a contract of 10,000 units of this good. The good costs $10 to produce, including a normal economic return on the capital employed and on managerial time. The company which offers the lowest price wins the contract; if the price is tied, the contract is shared between the two.

You could both charge $50, and divide up the contract. Each one would make $200,000 profit on it. This strategy is not, however, individually optimal. If your rival charges $50, you can reduce your price to $49.90, win the contract, and increase your profits to $399,000, leaving your rival out of the market. Clearly, the same thought occurs to your rival. At any price higher than $10, the two of you can improve your individual situation by undercutting the other and obtaining the entire contract.

Only when both of you are charging $10, and getting just a high enough return to make you stay in the market, is there no incentive for either of you to undercut the other. In the language of game theory, $10 is the only Nash Equilibrium of this game – the only price at which either of you cannot individually improve his own situation by reducing prices. But if the two of you charge $10, you are much worse off than you could have been had you shared the contract at $50.

Although the game Bertrand analysed is a simplified version of the pricing problem confronted by companies in markets with few competitors, the logic of the example still applies when some of its assumptions are weakened. It is generally the case that the individual incentive to cut prices by each participant in the industry leads to prices that leave each participant collectively worse off. For example, if two competitors sell products that are imperfect substitutes, undercutting one's rival by a small amount will not steal the entire market. This mitigates the incentive to cut price and avoids the stark results of zero profits, but still leads to prices below levels that maximise the joint profits of the two companies.

## Price reactions

The most important oversimplification of the Bertrand model of price competition is its one-shot, static set-up. In fact, virtually all price competition includes the possibility to react to a competitor's price decisions. A simple example of two adjacent retail petrol stations in a town illustrates the basic dynamics. For simplicity, we will assume there are no other petrol stations in town and that the town is somewhat isolated. To customers, one station's petrol is a perfect substitute for the other, so that if prices are unequal, the low-priced station will attract all the demand. Assume that both stations initially charge the full monopoly price (that price which maximises the joint profits of the two stations). What happens, if Station A, following Bertrand's logic, lowers its price by one cent per gallon? The manager of Station B will observe the price change as soon as his rival manager gets out a ladder and changes the sign. Knowing that this small price difference will reduce sales to zero, Station B's manager will surely respond by matching the price cut. Station A's manager, anticipating this response, should not lower his price in the first place. The outcome is the exact opposite of Bertrand's result – full monopoly pricing, rather than fully competitive pricing.

We have oversimplified but the example does demonstrate a fundamental logic of price competition. A company considering a price cut should bear in mind the entire impact of the price cut on profits. This includes the impact of any anticipated competitive response. If a company can anticipate price matching or even a more aggressive punishment by a rival, a price cut may be unprofitable.

Questions the potential price cutter must answer include: by how much can I increase market share before rivals respond, how will rivals respond, and what impact will the post-response set of prices have on my profits? The petrol station example involves a set of assumptions that make it easy to answer these questions and so predict a co-operative outcome. Among such assumptions are: price starts at the monopoly level, both competitors implicitly agree on what that level is, each observe the other's price immediately and without cost, consumers observe both prices simultaneously and at no cost, there are only two petrol stations and no relevant substitutes, and price changes are inexpensive.

However, most markets have more complex features than this gas station example making prediction more difficult; the key is understanding how to answer our basic questions in more complex environments. We now explore these issues.

**Industry conditions that affect the degree of price competition**

If all companies in an industry could get together and negotiate binding contracts setting each company's prices and production and also including payments across companies, it is possible they could reach an agreement that maximised industry profits. Such an agreement might be difficult to achieve as each company angled for a greater share of the profits. Of course, no court would enforce such an agreement, the very discussion of which probably violates antitrust laws.

The problem of co-ordinating at monopoly prices without violating antitrust laws is much more difficult, although a dynamic strategy such as tit-for-tat could achieve this outcome. The co-ordination process may be especially difficult when the objective is a price increase rather than prevention of a price cut. This is seen in the back and forth behaviour of the airlines after an attempt at raising prices (see Table 1 on Page 4). We discuss here some of the factors that will make implicitly co-operative outcomes through the use of variants of such strategies, more likely.

**1. The number of companies in an industry**

Co-operative pricing becomes more difficult as the number of companies increases. Co-ordination becomes more complex and less likely. Imagine the problems of co-ordinating price and quantity choices among many competing companies when communication is severely limited by antitrust laws.

The presence of many companies may lead to reduced incentive to punish a price cut. A company that cannot achieve a large market share may be able to avoid a response because the impact on other companies may be small relative to margins lost from responding. Also. observation of a rival's prices may become more difficult as the number of companies grows. As we explain below differences among companies are likely to increase as the number of companies rises. An example of the difficulty of maintaining cooperation with large numbers of players is the Organisation of Petroleum Exporting Countries, the oil cartel. Despite being able to raise prices and assign quotas in a manner that would be illegal if undertaken by individual producers in a country, OPEC has on each occasion finally failed to maintain high prices, in spite of occasional periods of success, such as the current one. One reason is that for any individual producer nation, increasing the amount of oil it pumps beyond its quota is profitable, as the reduction in its own revenues on the units it has already sold (not the revenues of the cartel, which do not enter this calculation) is more than offset by the increase in revenue due to the increase in the number of barrels sold.

**2. Differences among the companies in an industry**

Unlike our two petrol stations, no companies are exactly like their rivals. Companies differ in cost structure, capacity, product quality, existing

**continued on page 4**

## This week

**The dynamics of price competition**
Pricing has an important strategic role in economics say **Luis Garicano** and **Robert Gertner,** who go on to focus on the incentives companies need to use price as a competitive weapon.
**Pages 2-4**

**Alliances: achieving long-term value and short-term goals**
Alliances are risky but are a fundamental necessity of both short- and long-term strategy, says **Will Mitchell,** who also warns that unlike marriages, alliances are not for ever.
**Pages 6-7**

**Five ways to grow the market and create value**
With tasks like restructuring and downsizing behind them companies now need to concentrate on growth and value creation, suggests **Aneel G. Karnani.** Driving the pressure are shareholders, competitors and employees.
**Pages 8-10**

**Innovation beyond ideas: expectations in managing technology**
**Ron Adner** says that one of the toughest decisions a company can face is choosing innovative ideas that will both dominate existing markets and open up new ones.
**Pages 12-13**

**Challenges of running a regulated business**
Regulated businesses face most of the strategy issues that confront conventional companies, says **John Kay.** But they have others that are specific to their environment.
**Pages 14-15**

**COVER ILLUSTRATION**
Darren Raven

# The prisoner's dilemma in practice: tit-for-tat in the First World War trenches

Corbis

The 'price war game' is an example of a general game known as a ***prisoner's dilemma.*** In these games, the individual incentive of each player to defect (in the pricing game, to reduce prices) leads both players to a situation that leaves them collectively worse off. The original story involves two criminals who are arrested after committing a serious crime. The police have no proof of their involvement except for a minor infraction. The prosecutor offers them a deal whereby the one who implicates the other escapes all punishment and the other gets a heavy prison sentence. If both implicate the other, both end up in prison for a long time. The dominant strategy is for each criminal to implicate the other. Consequently they both end up getting relatively heavy prison sentences.

When the game is repeated, however, co-operation can be sustained implicitly through dynamic strategies. In his classic book, ***The Evolution of Co-operation,*** University of Michigan political scientist Robert Axelrod studied empirically and experimentally the strategies that lead players involved in prisoner's dilemma type situations to co-operative outcomes. His starting point was an unexpected experimental result. When experts were asked to submit strategies for repeated prisoner's dilemma games and these strategies were matched with each other in a computer tournament, ***tit-for-tat,*** the simplest strategy, won. This was a strategy that simply started co-operating and then did to the other player what that player had done to it previously.

Axelrod's analysis of the data found that tit-for-tat had four properties making a strategy successful. A successful strategy should be ***nice:*** confronted with a co-operative player, it should reciprocate. It should also be provocable: faced with an uncalled defection, it should respond. It should be forgiving: after responding to a defection, it should go back to co-operation. And it should be ***easy to understand:*** other players should be able to anticipate the consequences of their actions.

Axelrod presents a surprising example of the usefulness of a variant of this strategy: first world war trench warfare.

**Here is a summary of his account:**

"The historical situation in the quiet sectors along the Western Front was a [repeated] prisoner's dilemma. At any time, the choices of two small units facing each other are to shoot to kill or deliberately to shoot to avoid causing damage. For both sides, weakening the enemy is important because it promotes survival. Therefore, in the short run it is better to do damage now whether the enemy is shooting back or not.

"What made trench warfare so different from most other combat was that the same small units faced each other in immobile sectors for extended periods of time. This changed the game from a one move prisoner's dilemma in which defection is the dominant choice, to an iterated prisoner's dilemma in which conditional strategies are possible. The result accorded with the theory's predictions: ***with sustained interaction, the stable outcome could be mutual co-operation based upon reciprocity*** (emphasis added). In particular, both sides followed strategies that would not be the first to defect, but that would be provoked if the other defected.

"As the lines stabilised, non-aggression between the troops emerged spontaneously in many places along the front. The earliest instances may have been associated with meals which were served at the same times on both sides of no-man's land. An eyewitness noted that: 'In one section the hour of 8 to 9 a.m. was regarded as consecrated to private business, and certain places indicated by a flag were regarded as out of bounds by the snipers on both sides'. In the summer of 1915, [a soldier noted that]'It would be child's play to shell the road behind the enemy's trenches, crowded as it must be with ration wagons and water carts, into a bloodstained wilderness but on the whole there is silence. After all, if you prevent your enemy from drawing his rations, his remedy is simple: he will prevent you from drawing yours

"The strategies were provocable. During the periods of mutual restraint, the enemy soldiers took pains to show to each other that they could indeed retaliate if necessary. For example, German snipers showed their prowess to the British by aiming at spots on the walls of cottages and firing until they had cut a hole." (Extracted from Chapter 4 of ***The Evolution of Co-operation,*** by Robert Axelrod)

**Table 1. A variant of tit-for-tat in the airline industry in 1998**

| Date | Action |
|---|---|
| • March | Northwest begins sale on 14-day advance purchase fares. Others match. |
| • April | Continental tries to raise leisure fares $20 round-trip twice. Other carriers match, but Northwest blocks both attempts. |
| • May | Northwest initiates 25% sale on 21-day advance-purchase fares. Others match. Later, Continental tries a 5% increase on non-sale leisure fares. United, American, Delta and USAir all match, but not Northwest. Increase is rolled back. |
| • July | Continental raises leisure fares $20 round-trip; Northwest blocks. |
| August 11 | Delta and American raise fares by 4%, Northwest blocks. |
| August 14 | American raises fares by 4% in markets not served by Northwest; carriers other than Northwest match. |
| August 17 | Delta rolls back the increases; other airlines follow. |
| August 18 | Northwest raises fares by 4%. All other carriers match the increase |
| August 19 | Northwest rescinds fare hike; US Airways and TWA match same day. |
| August 20 | American, Delta, Continental match reversal; other carriers follow. |
| August 21 | Northwest hikes fares by 4%, other carriers match. |

Source: PaineWebber Airline Research, WSJ, authors elaboration.

## Price competition in internet retailing

**Internet retailing is a new and fast growing marketplace characterised by shopping convenience and low entry costs. The benefit to consumers from better information, reduced search costs, and access to a wider variety of goods and services is dramatic. This benefit for consumers need not necessarily translate, however, into higher long-run profitability for firms. Whether it does or not depends on the extent of price competition.**

**On a first approximation, it would appear that internet retailing is like our petrol station example – consumers can easily shift to a low-priced seller and transaction prices are transparent to competitors. This suggests pricing competition may be limited. A small number of similar firms competing for sales on the information superhighway would probably compete on price less aggressively than the same number of firms competing on an asphalt highway, as they would expect their price cuts to be immediately matched by their rivals, with no gain in market share.**

**This is not the whole story, however. Entry costs may be very low. It is possible to access large numbers of potential customers at a small fraction of the costs of telemarketing or the mail system and it may be possible to find and target specific segments of the population without the purchase of expensive lists. This should increase the number of firms and the differences among them, making high prices much more difficult to sustain. A small entrant with no brand name or reputation and few existing customers will find it necessary to compete on price to get a toehold in the market.**

**Moreover the reduction in search costs and the relative ease with which shoppers can compare prices on the internet would encourage consumers to switch to the low cost producers. In fact, the cost of switching in comparing is so low that there is a real possibility of negotiated prices being the norm. It may be much easier for a consumer to play competitors off against one another when one can get price quotes through electronic mail and make offers and counteroffers among a large number of sellers at very low cost. For example, playing one car dealer off against another is already quite easy thanks to the internet.**

## Most-favoured-customer clauses and price competition

**When a firm uses a most-favoured-customer clause (MFC), it promises each customer to give him or her the best price that the firm gives to anyone. The impact of such rules seems to be favourable to consumers, as each consumer gets the best possible treatment. In fact, such clauses decrease price competition, by making it very costly for firms to give discounts to their customers.Suppose a firm is negotiating a price with a particularly large customer. In the absence of an MFC, the firm may be tempted to reduce prices in order to gain this additional business. Now suppose the firm has an MFC clause, this means that if it gives a discount to this customer it will have to cut the prices to everyone. The trade-off now involved in reducing prices is cutting prices to all customers and decreasing the margin on all of the units sold in order to increase the quantity sold to this particular customer. Clearly, the incentives for the seller to cut prices are now smaller, making it likely that the prices paid by every consumer will increase.**

**In a 1997 study published in the Rand Journal of Economics, Yale School of Management Professor Fiona Scott Morton studied empirically the effect of a version of this clause introduced in the Medicare reimbursement program in 1990. The genesis of the change was the frustration of the US Congress in seeing that powerful health maintenance organisations obtained better prices from drug manufacturers than the government Medicaid program. In the 1990 Omnibus Budget Reconciliation Act, Congress legislated that manufacturers were required to sell branded drugs to Medicaid for the best price anyone paid for the drug or 87.5% of their average manufacturer price. The government expected prices to decrease as a result of the program. Scott Morton found that, in fact, the prices of branded drugs in competition with generics rose by 4% as a result of these new rules.**

**Why this unexpected effect? The program gives incentives to manufacturers to raise prices. As the director of drug purchasing of Kaiser Permanente, America's largest HMO put it (as quoted by Adam M. Brandenburger and Barry J. Nalebuff in their book, Co-opetition), "in the past we'd offer a drug manufacturer 90 percent of our business, maybe $10m in additional business, and get really low prices. Now, no one wants to go below the Medicaid floor." By adding an MFC provision to the law, the US Congress reduced the incentives of drug manufacturers to compete on prices, resulting in higher overall market prices.**

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market share, product characteristics, and product range. These differences make co-ordination more complex. The OPEC problem is not just a result of the number of member countries, but of differences in extraction costs, levels of reserves, wealth, refining interests, and political motives. Of course, as the number of companies increases, the differences among them are also likely to proliferate.

If companies differ they may not even agree on a single co-operative price - a low-cost company will prefer all companies to charge a lower price than a high-cost company. If companies differ, one company may find that it can cut price without inducing a response from rivals or one company may prefer low prices even if there is a competitive response. A new entrant into a market where market shares are quite stable may have an incentive to enter with low prices for either of these reasons. An incumbent may not respond because the loss in margin on all its customers may be greater than the small loss of share to the entrant. Even if all incumbents match the entrant's low price, the low price may attract new customers to the market. The entrant may get a disproportionate share of the new customers because they have no ties to the incumbents. An example of price-cutting by entrants is in the US wireless telephone market, where PCS companies, the first generation of digital wireless carriers, have entered with prices well below those of the two incumbent carriers.

### 3. The short run gain from cutting prices

To the extent that the benefit of undercutting high prices is small, the temptation to 'cheat' on an implicit high price agreement by lowering prices is also small. Two main factors may reduce the gains from short run price cuts: high capacity utilisation and high product differentiation. If little or no excess capacity exists, then little benefit in the form of higher sales can be derived from cutting prices. On the other hand, if a company has excess capacity there is a large incentive to reduce price and use additional capacity, particularly if the company enjoys a low marginal cost relative to prevailing prices. For example, the return to an airline from filling an empty seat or a hotel from filling an empty room, creates a strong incentive to cut price.

Similarly, a large degree of product differentiation between competing products also reduces the gains from undercutting the prices of rivals. If product differentiation is high, the increase in sales a company may derive from reducing its prices is small, as consumers of the other products are less likely to switch to a product they do not perceive as a close substitute to the one they consume.

### 4. Price transparency

In order to respond effectively to a rival's price cut, one must know that a price cut occurred. If secret price cuts are possible, matching of price cuts is slow or non-existent, and the incentive to cut prices is higher. The incentive to cut prices is smaller the more transparent the market, as price cuts are more easily matched in such a case. As a consequence, markets where prices or other sale conditions arise out of complex, private negotiations, are less likely to result in monopolistic pricing. Implicit co-ordination becomes especially difficult when companies cannot observe their rivals' competitive actions - a company must infer a rival's actions from market outcomes. It is only after losing many sales that you may be confident that a rival is pricing aggressively. The longer the price cuts go unpunished, the higher the incentive to cut prices.

There are market features that may affect the ability to infer rivals' actions. Frequent demand shocks, for example, make detecting price cuts by rivals more difficult. If one of the players observes a sudden decrease in sales, is he to conclude that the rival is undercutting his prices, or might it just be a "bad month" (or year)? Alternatively, if there is natural product or geographic segmentation, a company will learn if a competitor is trading aggressively by the loss of sales that "belong" to it.

To sum up, industries in which there are few companies, transparent prices and stable demand conditions, some easy way to implement tacit co-ordination and high product differentiation, are the most likely to be conducive to high price outcomes. But there is no simple formula for combining these elements to make predictions when different features work in different directions. Attempts to predict the extent of price competition in internet retailing offers an interesting case study of this problem. (See Price competition box above).

## Changing the Rules of the Game

Companies may embark on actions to change the rules of the game, by creating conditions that will lead to high prices.

The most common of these tactics usually involves making the market more transparent or decreasing the short run gains from price cuts, mainly through an increase in product differentiation. The antitrust treatment of such actions is very complex, so none of them should be taken as a strategic recommendation by the authors.

Companies have used many different tactics to increase market transparency. These include: attempts to create market segmentation, commitments to published price lists, advanced announcement of price changes, and price leadership by a company that, being larger than the rest, can play the price-setting role leaving all the other companies simply to imitate it.

Also, as we have argued, the short run gain from price cuts depends on the degree of product differentiation in the market. If differentiation increases, consumers are less likely to react to price cuts. An example of an action taken exclusively to increase the degree of product differentiation in a market, is the airline frequent flyer programme (FFP). If a customer is a member of the American Airlines FFP, he is less likely to respond to a low tariff offered by KLM on the same route. Thus KLM has a smaller incentive to cut prices, as do all other airlines. Higher ticket prices result.

Other ways exist to lower the temptation to cut prices by decreasing the short term gains from cutting prices. Among them is the use of most favoured customer clauses, which we discuss in the box above. ("Most-favoured-customer clauses and price competition").

## Signpost

Related articles can be found in Parts Three (FT Oct 11), Five (FT Oct 25) and Seven (FT Nov 8).